

JADAVPUR UNIVERSITY

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Department of Computer Science and Engineering, FET.

MASTER OF COMPUTER APPLICATION

1st Year 2nd Semester

DATABASE MANAGEMENT SYSTEM ASSIGNMENT

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Question 1. Create the following tables :

Tablename: EMPLOYEE

Structure :

EMP_CODE char(16)

EMP_NAME char(20)

DEPT_CODE char(16)

DESIG_CODE char(16)

SEX char(1)

ADDRESS char (25)

CITY char (20)

STATE char (20)

PIN char (6)

BASIC Number

JN_DT Date

Primary key is EMP_CODE

Tablename : DESIGNATION

Structure :

DESIG_CODE char(16)

DESIG_DESC char(20)

Primary key is DESIG_CODE

Tablename : DEPARTMENT

Structure :

DEPT_CODE char(16)

DEPT_NAME char(20)

Primary key is DEPT_CODE.

Query :

```
CREATE TABLE IF NOT EXISTS DEPARTMENT
( DEPT_CODE CHAR(16) PRIMARY KEY,
  DEPT_NAME CHAR(20)
);
```

```
CREATE TABLE IF NOT EXISTS DESIGNATION
( DESIG_CODE CHAR(16) PRIMARY KEY,
  DESIG_DESC CHAR(20)
);
```

```
CREATE TABLE IF NOT EXISTS EMPLOYEE
( EMP_CODE CHAR(16) PRIMARY KEY,
  EMP_NAME CHAR(20),
  DEPT_CODE CHAR(16),
  DESIG_CODE CHAR(16),
  SEX CHAR(1),
  ADDRESS CHAR(25),
  CITY CHAR(20),
  STATE CHAR(20),
  PIN CHAR(6),
  BASIC DECIMAL(10,2),
  JN_DT DATE,

  FOREIGN KEY (DEPT_CODE) REFERENCES DEPARTMENT(DEPT_CODE),
  FOREIGN KEY (DESIG_CODE) REFERENCES DESIGNATION(DESIG_CODE)
);
```

Output :

```
mysql> CREATE TABLE DEPARTMENT (DEPT_CODE CHAR(16) PRIMARY KEY, DEPT_NAME CHAR(20));  
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE DESIGNATION (DESIG_CODE CHAR(16) PRIMARY KEY, DESIG_DESC CHAR(20));  
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE EMPLOYEE (  
->     EMP_CODE CHAR(16) PRIMARY KEY,  
->     EMP_NAME CHAR(20),  
->     DEPT_CODE CHAR(16),  
->     DESIG_CODE CHAR(16),  
->     SEX CHAR(1),  
->     ADDRESS CHAR(25),  
->     CITY CHAR(20),  
->     STATE CHAR(20),  
->     PIN CHAR(6),  
->     BASIC DECIMAL(10,2),  
->     JN_DT DATE,  
->  
->     FOREIGN KEY (DEPT_CODE) REFERENCES DEPARTMENT(DEPT_CODE),  
->     FOREIGN KEY (DESIG_CODE) REFERENCES DESIGNATION(DESIG_CODE)  
-> );
```

```
Query OK, 0 rows affected (0.05 sec)
```

Question 2 : Display the structure of each table .

Query :

```
DESC department;  
DESC designation;  
DESC employee;
```

Output :

```
mysql> DESC department;
```

Field	Type	Null	Key	Default	Extra
DEPT_CODE	char(16)	NO	PRI	NULL	
DEPT_NAME	char(20)	YES		NULL	

```
2 rows in set (0.02 sec)
```

```
mysql> DESC designation;
```

Field	Type	Null	Key	Default	Extra
DESIG_CODE	char(16)	NO	PRI	NULL	
DESIG_DESC	char(20)	YES		NULL	

```
2 rows in set (0.00 sec)
```

```
mysql> DESC employee;
```

Field	Type	Null	Key	Default	Extra
EMP_CODE	char(16)	NO	PRI	NULL	
EMP_NAME	char(20)	YES		NULL	
DEPT_CODE	char(16)	YES	MUL	NULL	
DESIG_CODE	char(16)	YES	MUL	NULL	
SEX	char(1)	YES		NULL	
ADDRESS	char(25)	YES		NULL	
CITY	char(20)	YES		NULL	
STATE	char(20)	YES		NULL	
PIN	char(6)	YES		NULL	
BASIC	decimal(10,2)	YES		NULL	
JN_DT	date	YES		NULL	

```
11 rows in set (0.00 sec)
```

Question 3: Insert few rows in each table .

[While entering data in EMP table use DESIG_CODE which exists in DESIGNATION table and DEPT_CODE which exists in DEPARTMENT table. In DESIGNATION table, assign code for Manager, Executive, officer, clerk and helper. In DEPARTMENT table, assign code for Personnel, Production, Purchase, Finance, Research departments]

Query :

```
INSERT INTO DESIGNATION VALUES
```

```
('D001','Manager'),
('D002','Executive'),
('D003','Officer'),
('D004','Clerk'),
('D005','Helper');
```

```
INSERT INTO DEPARTMENT VALUES
```

```
('DP001','Personnel'),
('DP002','Production'),
('DP003','Purchase'),
('DP004','Finance'),
('DP005','Research');
```

```
INSERT INTO EMPLOYEE VALUES
```

```
('E001','Rahul Das','DP001','D001','M','12 Park
Street','Kolkata','WB','700001',60000,'2023-01-15'),
('E002','Anita Roy','DP002','D002','F','45 MG
Road','Kolkata','WB','700002',45000,'2023-02-10'),
('E003','Suman Sen','DP003','D003','M','21 Lake
Road','Siliguri','WB','734001',35000,'2022-07-20'),
('E004','Priya Dey','DP004','D004','F','10 College
St','Kolkata','WB','700073',25000,'2024-03-05'),
('E005','Amit Paul','DP005','D005','M','7 Hill Cart
Rd','Jalpaiguri','WB','735101',18000,'2024-04-12');
```

Output :

```
mysql> INSERT INTO DEPARTMENT VALUES
```

```
-> ('DP001','Personnel'),
-> ('DP002','Production'),
-> ('DP003','Purchase'),
-> ('DP004','Finance'),
-> ('DP005','Research');
```

```
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO DESIGNATION VALUES
```

```
-> ('D001','Manager'),
-> ('D002','Executive'),
-> ('D003','Officer'),
-> ('D004','Clerk'),
-> ('D005','Helper');
```

```
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO EMPLOYEE VALUES
```

```
-> ('E001','Rahul Das','DP001','D001','M','12 Park Street','Kolkata','WB','700001',60000,'2023-01-15'),
-> ('E002','Anita Roy','DP002','D002','F','45 MG Road','Kolkata','WB','700002',45000,'2023-02-10'),
-> ('E003','Suman Sen','DP003','D003','M','21 Lake Road','Siliguri','WB','734001',35000,'2022-07-20'),
-> ('E004','Priya Dey','DP004','D004','F','10 College St','Kolkata','WB','700073',25000,'2024-03-05'),
-> ('E005','Amit Paul','DP005','D005','M','7 Hill Cart Rd','Jalpaiguri','WB','735101',18000,'2024-04-12');
```

```
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

Question 4: In EMP table insert few rows without DEPT_CODE and BASIC.

Query :

```
INSERT INTO EMPLOYEE
(EMP_CODE,EMP_NAME,DESIG_CODE,SEX,ADDRESS,CITY,STATE,PIN,JN_DT) VALUES
('E006','Rita Ghosh','D003','F','22 Station Rd','Malda','WB','732101','2024-05-01'),
('E007','Arjun Roy','D004','M','15 Market Rd','Cooch Behar','WB','736101','2024-06-15');
```

Ouput :

```
mysql> INSERT INTO EMPLOYEE (EMP_CODE,EMP_NAME,DESIG_CODE,SEX,ADDRESS,CITY,STATE,PIN,JN_DT) VALUES
-> ('E006','Rita Ghosh','D003','F','22 Station Rd','Malda','WB','732101','2024-05-01'),
-> ('E007','Arjun Roy','D004','M','15 Market Rd','Cooch Behar','WB','736101','2024-06-15');
Query OK, 2 rows affected (0.00 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

Question 5: Find the rows with unassigned DEPT_CODE

Query :

```
SELECT * FROM EMPLOYEE WHERE DEPT_CODE IS NULL;
```

Output :

```
mysql> SELECT *
-> FROM EMPLOYEE
-> WHERE DEPT_CODE IS NULL;
```

EMP_CODE	EMP_NAME	DEPT_CODE	DESIG_CODE	SEX	ADDRESS	CITY	STATE	PIN	BASIC	JN_DT
E006	Rita Ghosh	NULL	D003	F	22 Station Rd	Malda	WB	732101	NULL	2024-05-01
E007	Arjun Roy	NULL	D004	M	15 Market Rd	Cooch Behar	WB	736101	NULL	2024-06-15

2 rows in set (0.00 sec)

Question 6 : Find the rows with BASIC equal to zero

Query :

```
SELECT * FROM EMPLOYEE WHERE BASIC = 0;
```

Ouput :

```
mysql> SELECT *
-> FROM EMPLOYEE
-> WHERE BASIC = 0;
Empty set (0.00 sec)
```

Question 7 : Find the rows with unassigned Basic [note down the difference between the results obtained in Q.6 and Q.7]

Query :

```
SELECT * FROM EMPLOYEE WHERE BASIC IS NULL;
```

Output :

```
mysql> SELECT *
-> FROM EMPLOYEE
-> WHERE BASIC IS NULL;
```

EMP_CODE	EMP_NAME	DEPT_CODE	DESIG_CODE	SEX	ADDRESS	CITY	STATE	PIN	BASIC	JN_DT
E006	Rita Ghosh	NULL	D003	F	22 Station Rd	Malda	WB	732101	NULL	2024-05-01
E007	Arjun Roy	NULL	D004	M	15 Market Rd	Cooch Behar	WB	736101	NULL	2024-06-15

2 rows in set (0.00 sec)

Question 8: Find the average basic of the employees.

Query :

```
SELECT AVG(BASIC) AS Average_Basic FROM EMPLOYEE;
```

Output :

```
mysql> SELECT AVG(BASIC) AS Average_Basic
-> FROM EMPLOYEE;

+-----+
| Average_Basic |
+-----+
| 36600.000000 |
+-----+
1 row in set (0.01 sec)
```

Question 9: Replace the BASIC with 0 for the rows with unassigned Basic.

Query :

```
UPDATE EMPLOYEE SET BASIC = 0 WHERE BASIC IS NULL;
```

Ouput :

```
mysql> UPDATE EMPLOYEE
-> SET BASIC = 0
-> WHERE BASIC IS NULL;
Query OK, 2 rows affected (0.02 sec)
Rows matched: 2 Changed: 2 Warnings: 0
```

Question 10 : Again, find the average Basic. (Note the difference of result obtained in Q.8 & Q.10.)

Query :

```
SELECT AVG(BASIC) AS Average_Basic FROM EMPLOYEE;
```

Output :

```
mysql> SELECT AVG(BASIC) AS Average_Basic
-> FROM EMPLOYEE;

+-----+
| Average_Basic |
+-----+
| 26142.857143 |
+-----+
1 row in set (0.00 sec)
```

Question 11 : Delete the rows with unassigned DEPT_CODE.

Query :

```
DELETE FROM EMPLOYEE WHERE DEPT_CODE IS NULL;
```

Output :

```
mysql> DELETE FROM EMPLOYEE
-> WHERE DEPT_CODE IS NULL;
Query OK, 2 rows affected (0.02 sec)
```

Question 12 : Say, Net pay of an employee= Basic+HRA+DA where HRA is 50% of the Basic & DA is 40% of Basic. Show the employee name & Net pay for all employees.

Query :

```
SELECT EMP_NAME, BASIC + (0.5 * BASIC) + (0.4 * BASIC) AS NET_PAY FROM EMPLOYEE;
```

Output :

```
mysql> SELECT EMP_NAME, BASIC + (0.5 * BASIC) + (0.4 * BASIC) AS NET_PAY FROM EMPLOYEE;
```

EMP_NAME	NET_PAY
Rahul Das	114000.000
Anita Roy	85500.000
Suman Sen	66500.000
Priya Dey	47500.000
Amit Paul	34200.000

5 rows in set (0.01 sec)

Question 13 : Show the EMP_NAME & BASIC in the ascending order of DEPT_CODE. The employee name must appear in uppercase.

Query :

```
SELECT UPPER(EMP_NAME) AS EMP_NAME, BASIC FROM EMPLOYEE ORDER BY DEPT_CODE ASC;
```

Output :

```
mysql> SELECT UPPER(EMP_NAME) AS EMP_NAME,  
-> BASIC FROM EMPLOYEE  
-> ORDER BY DEPT_CODE ASC;
```

EMP_NAME	BASIC
RAHUL DAS	60000.00
ANITA ROY	45000.00
SUMAN SEN	35000.00
PRIYA DEY	25000.00
AMIT PAUL	18000.00

5 rows in set (0.01 sec)

Question 14 : Find the employees who have joined after 1st January 2010.

Query :

```
SELECT * FROM EMPLOYEE WHERE JN_DT > '2010-01-01';
```

Output :

```
mysql> SELECT *  
-> FROM EMPLOYEE  
-> WHERE JN_DT > '2010-01-01';
```

EMP_CODE	EMP_NAME	DEPT_CODE	DESIG_CODE	SEX	ADDRESS	CITY	STATE	PIN	BASIC	JN_DT
E001	Rahul Das	DP001	D001	M	12 Park Street	Kolkata	WB	700001	60000.00	2023-01-15
E002	Anita Roy	DP002	D002	F	45 MG Road	Kolkata	WB	700002	45000.00	2023-02-10
E003	Suman Sen	DP003	D003	M	21 Lake Road	Siliguri	WB	734001	35000.00	2022-07-20
E004	Priya Dey	DP004	D004	F	10 College St	Kolkata	WB	700073	25000.00	2024-03-05
E005	Amit Paul	DP005	D005	M	7 Hill Cart Rd	Jalpaiguri	WB	735101	18000.00	2024-04-12

5 rows in set (0.01 sec)

Question 15: Find, how many employees have joined in the month of January?

Query :

```
SELECT COUNT(*) AS JAN_JOINED FROM EMPLOYEE WHERE MONTH(JN_DT) = 1;
```

Output :

```
mysql> SELECT COUNT(*) AS JAN_JOINED
-> FROM EMPLOYEE
-> WHERE MONTH(JN_DT) = 1;

+-----+
| JAN_JOINED |
+-----+
|          1 |
+-----+
1 row in set (0.01 sec)
```

Question 16 : Find the maximum & minimum Basic.

Query :

```
SELECT MAX(BASIC) AS MAX_BASIC, MIN(BASIC) AS MIN_BASIC FROM EMPLOYEE;
```

Output :

```
mysql> SELECT MAX(BASIC) AS MAX_BASIC, MIN(BASIC) AS MIN_BASIC FROM EMPLOYEE;

+-----+-----+
| MAX_BASIC | MIN_BASIC |
+-----+-----+
| 60000.00 | 18000.00 |
+-----+-----+
1 row in set (0.00 sec)
```

Question 17 : Find how many Female employees are there?

Query :

```
SELECT COUNT(*) AS FEMALE_EMPLOYEES FROM EMPLOYEE WHERE SEX = 'F';
```

Output :

```
mysql> SELECT COUNT(*) AS FEMALE_EMPLOYEES FROM EMPLOYEE WHERE SEX = 'F';

+-----+
| FEMALE_EMPLOYEES |
+-----+
|                2 |
+-----+
1 row in set (0.01 sec)
```

Question 18: Replace CITY with existing value converted into uppercase for all rows.

Query :

```
UPDATE EMPLOYEE SET CITY = UPPER(CITY);
```

Output :

```
mysql> UPDATE EMPLOYEE SET CITY = UPPER(CITY);
Query OK, 5 rows affected (0.01 sec)
Rows matched: 5  Changed: 5  Warnings: 0
```

Question 19: Find in how many different cities various employees are residing?

Query :

```
SELECT COUNT(DISTINCT CITY) AS TOTAL_CITIES FROM EMPLOYEE;
```

Output :

```
mysql> SELECT COUNT(DISTINCT CITY) AS TOTAL_CITIES FROM EMPLOYEE;
```

```
+-----+
| TOTAL_CITIES |
+-----+
|           3 |
+-----+
```

```
1 row in set (0.01 sec)
```

Question 20 : Display the employee information in the ascending order of DEPT_CODE and with in a Department, it should be in the descending order of BASIC

Query :

```
SELECT * FROM EMPLOYEE ORDER BY DEPT_CODE ASC, BASIC DESC;
```

Output :

```
mysql> SELECT * FROM EMPLOYEE ORDER BY DEPT_CODE ASC, BASIC DESC;
```

EMP_CODE	EMP_NAME	DEPT_CODE	DESIG_CODE	SEX	ADDRESS	CITY	STATE	PIN	BASIC	JN_DT
E001	Rahul Das	DP001	D001	M	12 Park Street	KOLKATA	WB	700001	60000.00	2023-01-15
E002	Anita Roy	DP002	D002	F	45 MG Road	KOLKATA	WB	700002	45000.00	2023-02-10
E003	Suman Sen	DP003	D003	M	21 Lake Road	SILIGURI	WB	734001	35000.00	2022-07-20
E004	Priya Dey	DP004	D004	F	10 College St	KOLKATA	WB	700073	25000.00	2024-03-05
E005	Amit Paul	DP005	D005	M	7 Hill Cart Rd	JALPAIGURI	WB	735101	18000.00	2024-04-12

```
5 rows in set (0.00 sec)
```

Question 21. From the EMP table show the minimum, maximum and average basic for each department (show dept. Code).

Query :

```
SELECT
    DEPT_CODE,
    MIN(BASIC) AS MIN_BASIC,
    MAX(BASIC) AS MAX_BASIC,
    AVG(BASIC) AS AVG_BASIC
FROM EMPLOYEE
GROUP BY DEPT_CODE;
```

```
mysql> SELECT
->     DEPT_CODE,
->     MIN(BASIC) AS MIN_BASIC,
->     MAX(BASIC) AS MAX_BASIC,
->     AVG(BASIC) AS AVG_BASIC
-> FROM EMPLOYEE
-> GROUP BY DEPT_CODE;
```

DEPT_CODE	MIN_BASIC	MAX_BASIC	AVG_BASIC
DP001	19106.00	89977.00	55118.742574
DP002	19451.00	89732.00	54488.371859
DP003	18230.00	89438.00	53906.551724
DP004	18251.00	89935.00	54143.875598
DP005	18000.00	89914.00	55452.692708

5 rows in set (0.06 sec)

Question 22. Find the number of female employees in each department (show dept. Code).

Query :

```
SELECT
    DEPT_CODE,
    COUNT(*) AS FEMALE_EMPLOYEES
FROM EMPLOYEE
WHERE SEX='F'
GROUP BY DEPT_CODE;
```

```
mysql> SELECT
->     DEPT_CODE,
->     COUNT(*) AS FEMALE_EMPLOYEES
-> FROM EMPLOYEE
-> WHERE SEX='F'
-> GROUP BY DEPT_CODE;
```

DEPT_CODE	FEMALE_EMPLOYEES
DP001	94
DP002	97
DP003	98
DP004	119
DP005	96

5 rows in set (0.00 sec)

Question 23: Find the city wise no. of employees for each department (show dept. Code).

Query :

```
SELECT
    DEPT_CODE,
    CITY,
    COUNT(*) AS TOTAL_EMPLOYEES
FROM EMPLOYEE
GROUP BY DEPT_CODE, CITY;
```

```
mysql> SELECT
->     DEPT_CODE,
->     CITY,
->     COUNT(*) AS TOTAL_EMPLOYEES
-> FROM EMPLOYEE
-> GROUP BY DEPT_CODE, CITY;
```

DEPT_CODE	CITY	TOTAL_EMPLOYEES
DP002	Guwahati	3
DP001	Murshidabad	4
DP001	KOLKATA	11
DP005	Kharagpur	6
DP002	Kolkata	3
DP002	Agra	2
DP001	Chandigarh	9
DP005	Patna	3
DP004	Faridabad	6
DP002	Bardhaman	7
DP004	Gurugram	6
DP002	Pune	7
DP004	Chandigarh	4
DP005	Madurai	5
DP004	Mumbai	4
DP002	Jalpaiguri	1

Question 24 : Show the designation wise no of employees who have joined in the year 2000 in each department. The listing should appear in the ascending order of no. of employees.

Query :

```
SELECT
    E.DEPT_CODE,
    D.DESIG_DESC, COUNT(*)
    AS EMP_COUNT
FROM EMPLOYEE E
JOIN DESIGNATION D
    ON E.DESIG_CODE=D.DESIG_CODE
WHERE YEAR(E.JN_DT)=2000
GROUP BY E.DEPT_CODE,D.DESIG_DESC
ORDER BY EMP_COUNT ASC;
```

```
mysql> SELECT
->     E.DEPT_CODE,
->     D.DESIG_DESC,
->     COUNT(*) AS EMP_COUNT
-> FROM EMPLOYEE E
-> JOIN DESIGNATION D
->     ON E.DESIG_CODE=D.DESIG_CODE
-> WHERE YEAR(E.JN_DT)=2000
-> GROUP BY E.DEPT_CODE,D.DESIG_DESC
-> ORDER BY EMP_COUNT ASC;
```

DEPT_CODE	DESIG_DESC	EMP_COUNT
DP002	Manager	1
DP002	Executive	1
DP001	Executive	1
DP003	Officer	1
DP001	Officer	1
DP002	Clerk	1
DP003	Clerk	1
DP002	Helper	1
DP001	Helper	1
DP004	Manager	2
DP001	Manager	2
DP005	Manager	2
DP003	Executive	2
DP002	Officer	2
DP005	Clerk	2
DP003	Manager	3
DP004	Executive	3
DP005	Officer	3
DP004	Clerk	3
DP003	Helper	3
DP004	Helper	4

21 rows in set (0.00 sec)

Question 25: Find the department code wise total basic of male employees only for the departments for which such total is more than 50,000 and the listing should appear in the descending order of total basic.

Query :

```
SELECT
    DEPT_CODE,
    SUM(BASIC) AS TOTAL_BASIC
FROM EMPLOYEE
WHERE SEX='M'
GROUP BY DEPT_CODE
HAVING SUM(BASIC) > 50000
ORDER BY TOTAL_BASIC DESC;
```

```
mysql> SELECT
->     DEPT_CODE,
->     SUM(BASIC) AS TOTAL_BASIC
-> FROM EMPLOYEE
-> WHERE SEX='M'
-> GROUP BY DEPT_CODE
-> HAVING SUM(BASIC) > 50000
-> ORDER BY TOTAL_BASIC DESC;
```

DEPT_CODE	TOTAL_BASIC
DP001	6224335.00
DP003	5635163.00
DP002	5564516.00
DP004	5089610.00
DP005	4952076.00

5 rows in set (0.00 sec)

Question 26: Show the employee name, Designation description and basic for all employees.

Query :

```
SELECT
    E.EMP_NAME,
    D.DESIG_DESC,
    E.BASIC
FROM EMPLOYEE E
JOIN DESIGNATION D
ON E.DESIG_CODE=D.DESIG_CODE;
```

```
mysql> SELECT
->     E.EMP_NAME,
->     D.DESIG_DESC,
->     E.BASIC
-> FROM EMPLOYEE E
-> JOIN DESIGNATION D
-> ON E.DESIG_CODE=D.DESIG_CODE;
```

EMP_NAME	DESIG_DESC	BASIC
Rahul Das	Manager	60000.00
Deep Khanna	Manager	61575.00
Biswajit Yadav	Manager	53196.00
Rahul Yadav	Manager	33396.00
Rakesh Menon	Manager	34600.00
Sudip Mitra	Manager	38001.00
Ankita Sharma	Manager	22993.00
Avijit Ganguly	Manager	24922.00
Poulomi Sarkar	Manager	82879.00
Santanu Pramanik	Manager	31511.00
Prasenjit Barman	Manager	54864.00
Priva Mehta	Manager	39760.00

Question 27 : Show the employee name, Designation description, Department Name & Basic for all employees.

Query :

```
SELECT
    E.EMP_NAME,
    DG.DESIG_DESC,
    DP.DEPT_NAME,
    E.BASIC
FROM EMPLOYEE E
JOIN DESIGNATION DG
    ON E.DESIG_CODE=DG.DESIG_CODE
JOIN DEPARTMENT DP
    ON E.DEPT_CODE=DP.DEPT_CODE;
```

```
mysql> SELECT
->     E.EMP_NAME,
->     DG.DESIG_DESC,
->     DP.DEPT_NAME,
->     E.BASIC
-> FROM EMPLOYEE E
-> JOIN DESIGNATION DG
->     ON E.DESIG_CODE=DG.DESIG_CODE
-> JOIN DEPARTMENT DP
->     ON E.DEPT_CODE=DP.DEPT_CODE;
```

EMP_NAME	DESIG_DESC	DEPT_NAME	BASIC
Gourav Poddar	Executive	Personnel	28212.00
Rahul Das	Manager	Personnel	60000.00
Akash Nandi	Executive	Personnel	65052.00
Madhumita Naidu	Executive	Personnel	54267.00
Rakesh Singh	Clerk	Personnel	61017.00
Bikash Gupta	Officer	Personnel	30660.00
Tania Debnath	Clerk	Personnel	20725.00
Debjani Poddar	Officer	Personnel	64025.00
Rakesh Menon	Manager	Personnel	34600.00
Rahul Nair	Executive	Personnel	50785.00

Question 28 : Find the department Codes in which no employee works.

Query :

```
SELECT DEPT_CODE
FROM DEPARTMENT
WHERE DEPT_CODE NOT IN
    ( SELECT DISTINCT
      DEPT_CODE FROM EMPLOYEE
      WHERE DEPT_CODE IS NOT NULL
    );
```

```
mysql> SELECT DEPT_CODE
-> FROM DEPARTMENT
-> WHERE DEPT_CODE NOT IN
-> (
->     SELECT DISTINCT DEPT_CODE
->     FROM EMPLOYEE
->     WHERE DEPT_CODE IS NOT NULL
-> );
```

```
+-----+
| DEPT_CODE |
+-----+
| DP006     |
+-----+
1 row in set (0.00 sec)
```

Question 29 : Find the department names where at least one employee works.

Query :

```
SELECT DISTINCT D.DEPT_NAME
FROM DEPARTMENT D JOIN EMPLOYEE E ON D.DEPT_CODE=E.DEPT_CODE;
```

```
mysql> SELECT DISTINCT D.DEPT_NAME
-> FROM DEPARTMENT D
-> JOIN EMPLOYEE E
-> ON D.DEPT_CODE=E.DEPT_CODE;
```

```
+-----+
| DEPT_NAME |
+-----+
| Personnel |
| Production |
| Purchase  |
| Finance   |
| Research  |
+-----+
```

5 rows in set (0.00 sec)

Question 30: Find the department names where at least 10 employees work.

Query :

```
SELECT D.DEPT_NAME
FROM DEPARTMENT D JOIN EMPLOYEE E ON D.DEPT_CODE=E.DEPT_CODE
GROUP BY D.DEPT_CODE,D.DEPT_NAME HAVING COUNT(*) >= 10;
```

```
mysql> SELECT
-> D.DEPT_NAME
-> FROM DEPARTMENT D
-> JOIN EMPLOYEE E
-> ON D.DEPT_CODE=E.DEPT_CODE
-> GROUP BY D.DEPT_CODE,D.DEPT_NAME
-> HAVING COUNT(*) >= 10;
```

```
+-----+
| DEPT_NAME |
+-----+
| Personnel |
| Production |
| Purchase  |
| Finance   |
| Research  |
+-----+
```

5 rows in set (0.00 sec)

Question 31: Find the department code in which employee with highest Basic works.

Query :

```
SELECT DEPT_CODE FROM EMPLOYEE WHERE BASIC =
( SELECT MAX(BASIC)
FROM EMPLOYEE
);
```

```
mysql> SELECT DEPT_CODE
-> FROM EMPLOYEE
-> WHERE BASIC =
-> (
-> SELECT MAX(BASIC)
-> FROM EMPLOYEE
-> );
```

```
+-----+
| DEPT_CODE |
+-----+
| DP001     |
+-----+
```

1 row in set (0.00 sec)

Question 32: Find the Designation description of the employee with highest basic.

Query :

```
SELECT D.DESIG_DESC
FROM EMPLOYEE E
JOIN DESIGNATION D
      ON E.DESIG_CODE=D.DESIG_CODE
WHERE E.BASIC =
(
  SELECT MAX(BASIC)
  FROM EMPLOYEE
);
```

```
mysql> SELECT D.DESIG_DESC
-> FROM EMPLOYEE E
-> JOIN DESIGNATION D
->      ON E.DESIG_CODE=D.DESIG_CODE
-> WHERE E.BASIC =
-> (
->      SELECT MAX(BASIC)
->      FROM EMPLOYEE
-> );
```

```
+-----+
| DESIG_DESC |
+-----+
| Executive  |
+-----+
```

1 row in set (0.00 sec)

Question 33: Find the no. of managers in each department.

Query :

```
SELECT
      E.DEPT_CODE,
      COUNT(*) AS TOTAL MANAGERS
FROM EMPLOYEE E
JOIN DESIGNATION D
      ON E.DESIG_CODE=D.DESIG_CODE
WHERE D.DESIG_DESC='Manager'
GROUP BY E.DEPT_CODE;
```

```
mysql> SELECT
->      E.DEPT_CODE,
->      COUNT(*) AS TOTAL MANAGERS
-> FROM EMPLOYEE E
-> JOIN DESIGNATION D
->      ON E.DESIG_CODE=D.DESIG_CODE
-> WHERE D.DESIG_DESC='Manager'
-> GROUP BY E.DEPT_CODE;
```

```
+-----+-----+
| DEPT_CODE | TOTAL MANAGERS |
+-----+-----+
| DP001     | 34              |
| DP002     | 38              |
| DP003     | 39              |
| DP005     | 30              |
| DP004     | 40              |
+-----+-----+
```

5 rows in set (0.00 sec)

Question 34: Find the maximum basic from EMP table without using MAX().

Query :

```
SELECT BASIC AS MAX_BASIC
FROM EMPLOYEE
ORDER BY BASIC DESC
LIMIT 1;
```

```
mysql> SELECT BASIC AS MAX_BASIC
-> FROM EMPLOYEE
-> ORDER BY BASIC DESC
-> LIMIT 1;
```

```
+-----+
```

```
| MAX_BASIC |
```

```
+-----+
```

```
| 89977.00 |
```

```
+-----+
```

```
1 row in set (0.00 sec)
```

Question 35: Find the minimum basic from EMP table without using MIN().

Query :

```
SELECT BASIC AS MIN_BASIC
FROM EMPLOYEE
ORDER BY BASIC ASC
LIMIT 1;
```

```
mysql> SELECT BASIC AS MIN_BASIC
-> FROM EMPLOYEE
-> ORDER BY BASIC ASC
-> LIMIT 1;
```

```
+-----+
```

```
| MIN_BASIC |
```

```
+-----+
```

```
| 18000.00 |
```

```
+-----+
```

```
1 row in set (0.00 sec)
```

Question 36: Find the name of the department with highest total basic. Do the same for highest average basic and maximum no. of employee.

Query :

```
SELECT D.DEPT_NAME
FROM DEPARTMENT D JOIN EMPLOYEE E ON D.DEPT_CODE=E.DEPT_CODE
GROUP BY D.DEPT_CODE,D.DEPT_NAME
ORDER BY SUM(E.BASIC) DESC LIMIT 1;
```

```
SELECT D.DEPT_NAME
FROM DEPARTMENT D JOIN EMPLOYEE E ON D.DEPT_CODE=E.DEPT_CODE
GROUP BY D.DEPT_CODE,D.DEPT_NAME
ORDER BY AVG(E.BASIC) DESC LIMIT 1;
```

```
SELECT D.DEPT_NAME
FROM DEPARTMENT D JOIN EMPLOYEE E ON D.DEPT_CODE=E.DEPT_CODE
GROUP BY D.DEPT_CODE,D.DEPT_NAME
ORDER BY COUNT(*) DESC LIMIT 1;
```

```
mysql> SELECT D.DEPT_NAME
-> FROM DEPARTMENT D
-> JOIN EMPLOYEE E
-> ON D.DEPT_CODE=E.DEPT_CODE
-> GROUP BY D.DEPT_CODE,D.DEPT_NAME
-> ORDER BY SUM(E.BASIC) DESC
-> LIMIT 1;
```

DEPT_NAME
Finance

1 row in set (0.00 sec)

```
mysql> SELECT D.DEPT_NAME
-> FROM DEPARTMENT D
-> JOIN EMPLOYEE E
-> ON D.DEPT_CODE=E.DEPT_CODE
-> GROUP BY D.DEPT_CODE,D.DEPT_NAME
-> ORDER BY AVG(E.BASIC) DESC
-> LIMIT 1;
```

DEPT_NAME
Research

1 row in set (0.00 sec)

```
mysql> SELECT D.DEPT_NAME
-> FROM DEPARTMENT D
-> JOIN EMPLOYEE E
-> ON D.DEPT_CODE=E.DEPT_CODE
-> GROUP BY D.DEPT_CODE,D.DEPT_NAME
-> ORDER BY COUNT(*) DESC
-> LIMIT 1;
```

DEPT_NAME
Finance

1 row in set (0.00 sec)

Question 37: Insert same rows into EMP table with designation code not existing in DESIGNATION table.

Query :

INSERT INTO EMPLOYEE

VALUES

('E100','Test User','DP001','D999','M',
'ABC Road','Kolkata','WB','700001',
30000,'2024-01-01');

```
mysql> INSERT INTO EMPLOYEE
-> VALUES
-> ('E100','Test User','DP001','D999','M',
-> 'ABC Road','Kolkata','WB','700001',
-> 30000,'2024-01-01');
```

ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails (`assignments`.`employee`, CONSTRAINT `employee\_ibfk\_2` FOREIGN KEY (`DESIG\_CODE`) REFERENCES `designation` (`DESIG\_CODE`))

After dropping the foreign key :

```
mysql> INSERT INTO EMPLOYEE
-> VALUES
-> ('E100','Test User','DP001','D999','M',
-> 'ABC Road','Kolkata','WB','700001',
-> 30000,'2024-01-01');
```

Query OK, 1 row affected (0.01 sec)

Question 38: Delete the rows from EMP table with invalid DESIG_CODE.

Query :

```
DELETE FROM EMPLOYEE WHERE DESIG_CODE NOT IN
( SELECT DESIG_CODE
  FROM DESIGNATION
);
```

```
mysql> DELETE FROM EMPLOYEE
-> WHERE DESIG_CODE NOT IN
-> (
->     SELECT DESIG_CODE
->     FROM DESIGNATION
-> );
Query OK, 0 rows affected (0.00 sec)
```

Question 39: Find the name of the female employees with basic greater than the average basic of their respective department.

Query :

```
SELECT E1.EMP_NAME FROM EMPLOYEE E1 WHERE E1.SEX='F' AND E1.BASIC >
( SELECT AVG(E2.BASIC)
  FROM EMPLOYEE E2
 WHERE E2.DEPT_CODE=E1.DEPT_CODE
);
```

```
mysql> SELECT
->     E1.EMP_NAME
-> FROM EMPLOYEE E1
-> WHERE E1.SEX='F'
-> AND E1.BASIC >
-> (
->     SELECT AVG(E2.BASIC)
->     FROM EMPLOYEE E2
->     WHERE E2.DEPT_CODE=E1.DEPT_CODE
-> );
```

EMP_NAME
Rohit Singh
Anirban Pillai
Akash Nandi
Pallavi Kapoor
Ritika Patel
Debjani Agarwal
Santanu Bera
Payel Debnath
Prasenjit Barman

Question 40: Find the number of female managers.

Query :

```
SELECT COUNT(*) AS FEMALE MANAGERS FROM EMPLOYEE E JOIN DESIGNATION D ON
E.DESIG_CODE=D.DESIG_CODE WHERE E.SEX='F' AND D.DESIG_DESC='Manager';
```

```
mysql> SELECT COUNT(*) AS FEMALE MANAGERS
-> FROM EMPLOYEE E
-> JOIN DESIGNATION D
-> ON E.DESIG_CODE=D.DESIG_CODE
-> WHERE E.SEX='F'
-> AND D.DESIG_DESC='Manager';
```

FEMALE MANAGERS
91

1 row in set (0.00 sec)

Question 41: In an organization, number of departments exists. Each department has a name & unique code. Number of employees work in each department. Each employee has unique employee code. Detailed information like name, address, city, basic, date of join are also stored. In a leave register for each employee leave records are kept showing leave type (CL/EL/ML etc.), from-date and to-date. When an employee retires or resigns then all the leave information pertaining to him are also deleted. Basic salary must be within Rs.5000 to Rs.9000. A department can not be deleted if any employee record refers to it. Valid grades are A/B/C. Employee name must be in uppercase only. Default value for joining date is system date. Design & implement the tables with necessary constraints to support the scenario depicted above.

Query :

```
CREATE TABLE Department(  
    dcode VARCHAR(10) PRIMARY KEY,  
    dname VARCHAR(50) NOT NULL UNIQUE,  
    no_of_employee INT DEFAULT 0  
);  
  
CREATE TABLE  
    Employee( empcode INT  
    PRIMARY KEY,  
    ename VARCHAR(50) NOT NULL,  
    address VARCHAR(100),  
    city VARCHAR(50),  
    basic DECIMAL(8,2),  
    doj DATE DEFAULT (CURRENT_DATE),  
    grade CHAR(1),  
    dcode VARCHAR(10),  
  
    CONSTRAINT chk_basic CHECK (basic BETWEEN 5000 AND 9000),  
    CONSTRAINT chk_grade CHECK (grade IN ('A','B','C')),  
    CONSTRAINT chk_uppercase CHECK (ename = UPPER(ename)),  
    CONSTRAINT fk_emp_dept FOREIGN KEY (dcode) REFERENCES Department(dcode) ON  
DELETE RESTRICT  
);  
  
CREATE TABLE LeaveRegister(  
    leave_id INT AUTO_INCREMENT PRIMARY KEY,  
    empcode INT NOT NULL,  
    leave_type VARCHAR(10) NOT NULL,  
    from_date DATE NOT NULL,  
    to_date DATE NOT NULL,  
  
    CONSTRAINT fk_leave_emp FOREIGN KEY(empcode) REFERENCES Employee(empcode)  
ON DELETE CASCADE  
);
```

Output :

```
mysql> CREATE TABLE Department(  
->    dcode VARCHAR(10) PRIMARY KEY,  
->    dname VARCHAR(50) NOT NULL UNIQUE,  
->    no_of_employee INT DEFAULT 0  
-> );  
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE Employee(  
->     empcode INT PRIMARY KEY,  
->     ename VARCHAR(50) NOT NULL,  
->     address VARCHAR(100),  
->     city VARCHAR(50),  
->     basic DECIMAL(8,2),  
->     doj DATE DEFAULT (CURRENT_DATE),  
->     grade CHAR(1),  
->     dcode VARCHAR(10),  
->  
->     CONSTRAINT chk_basic CHECK (basic BETWEEN 5000 AND 9000),  
->     CONSTRAINT chk_grade CHECK (grade IN ('A','B','C')),  
->     CONSTRAINT chk_uppercase CHECK (ename = UPPER(ename)),  
->     CONSTRAINT fk_emp_dept FOREIGN KEY (dcode) REFERENCES Department(dcode) ON DELETE RESTRICT  
-> );
```

Query OK, 0 rows affected (0.03 sec)

```
mysql> CREATE TABLE LeaveRegister(  
->     leave_id INT AUTO_INCREMENT PRIMARY KEY,  
->     empcode INT NOT NULL,  
->     leave_type VARCHAR(10) NOT NULL,  
->     from_date DATE NOT NULL,  
->     to_date DATE NOT NULL,  
->  
->     CONSTRAINT fk_leave_emp FOREIGN KEY(empcode) REFERENCES Employee(empcode) ON DELETE CASCADE  
-> );
```

Query OK, 0 rows affected (0.02 sec)

Question 42: Try to violate the constraints that you have implemented in the table & note, what happens. [Try with suitable INSERT/UPDATE/DELETE instruction]

Query :

```
INSERT INTO Employee(empcode,ename,address,city,basic,grade,dcode) VALUES
(104,'RAKESH','ABC','KOLKATA',10000,'A','D1');
```

```
INSERT INTO Employee (empcode,ename,address,city,basic,grade,dcode) VALUES
(105,'RAMESH','ABC','KOLKATA',7000,'D','D1');
```

```
INSERT INTO Employee(empcode,ename,address,city,basic,grade,dcode) VALUES
(107,'SOUMEN','ABC','KOLKATA',7000,'A','D10');
```

```
DELETE FROM Department
WHERE dcode='D1';
```

```
DELETE FROM Employee
WHERE empcode=101;
```

Output :

```
mysql> INSERT INTO Employee(empcode,ename,address,city,basic,grade,dcode) VALUES
-> (104,'RAKESH','ABC','KOLKATA',10000,'A','D1');
ERROR 3819 (HY000): Check constraint 'chk_basic' is violated.
```

```
mysql> INSERT INTO Employee (empcode,ename,address,city,basic,grade,dcode) VALUES
-> (105,'RAMESH','ABC','KOLKATA',7000,'D','D1');
ERROR 3819 (HY000): Check constraint 'chk_grade' is violated.
```

```
mysql> INSERT INTO Employee(empcode,ename,address,city,basic,grade,dcode) VALUES
-> (107,'SOUMEN','ABC','KOLKATA',7000,'A','D10');
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails
(`assignments`.`employee`, CONSTRAINT `fk_emp_dept` FOREIGN KEY (`dcode`) REFERENCES
`department` (`dcode`) ON DELETE RESTRICT)
```

```
mysql> DELETE FROM Department
-> WHERE dcode='D1';
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key
constraint fails (`assignments`.`employee`, CONSTRAINT `fk_emp_dept` FOR
EIGN KEY (`dcode`) REFERENCES `department` (`dcode`) ON DELETE RESTRICT)
mysql>
```

```
mysql> DELETE FROM Employee
-> WHERE empcode=101;
Query OK, 1 row affected (0.00 sec)
```

Question 43:

a) create a view showing employee code, name, dcode & Basic For a particular department.

Query :

```
CREATE VIEW D1_EMP_VIEW AS
```

```
SELECT empcode, ename, dcode, basic FROM Employee WHERE dcode='D1';
```

Output :

```
mysql> CREATE VIEW D1_EMP_VIEW AS
-> SELECT empcode, ename, dcode, basic FROM Employee WHERE dcode='D1';
Query OK, 0 rows affected (0.00 sec)
```

b) Try to ensure a row into the view with valid department & also with invalid ones.

Query :

```
INSERT INTO D1_EMP_VIEW VALUES(201, 'ANIK', 'D1', 7500);
```

```
INSERT INTO D1_EMP_VIEW VALUES(202, 'ROHIT', 'D5', 7000);
```

Output :

```
mysql> INSERT INTO D1_EMP_VIEW VALUES
-> (201, 'ANIK', 'D1', 7500);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO D1_EMP_VIEW VALUES
-> (202, 'ROHIT', 'D5', 7000);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails
(`assignments`.`employee`, CONSTRAINT `fk_emp_dept` FOREIGN KEY (`dcode`) REFERENCES
`department` (`dcode`) ON DELETE RESTRICT)
```

c) Find the newly inserted row in the table From which view was created .

Query :

```
SELECT * FROM Employee WHERE empcode=201;
```

Output :

```
mysql> SELECT * FROM Employee
-> WHERE empcode=201;

+-----+-----+-----+-----+-----+-----+-----+-----+
| empcode | ename | address | city | basic | doj | grade | dcode |
+-----+-----+-----+-----+-----+-----+-----+-----+
|      201 | ANIK  | NULL    | NULL | 7500.00 | NULL | NULL | D1    |
+-----+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

d) Try to increment basic by Rs.100/-

Query :

```
UPDATE D1_EMP_VIEW SET basic = basic + 100;
```

Output :

```
mysql> UPDATE D1_EMP_VIEW
-> SET basic = basic + 100;
Query OK, 2 rows affected (0.00 sec)
Rows matched: 2 Changed: 2 Warnings: 0
```

e) Check it in the original table.

Query :

```
SELECT empcode, ename, basic FROM Employee;
```

Output :

```
mysql> SELECT empcode, ename, basic
-> FROM Employee;

+-----+-----+-----+
| empcode | ename | basic |
+-----+-----+-----+
|      102 | AMIT  | 6600.00 |
|      103 | SUMAN | 8000.00 |
|      201 | ANIK  | 7600.00 |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

f) Delete the view.

Query :

```
DROP VIEW D1_EMP_VIEW;
```

Output :

```
mysql> DROP VIEW D1_EMP_VIEW;
Query OK, 0 rows affected (0.01 sec)
```

Question 44:

a) create a view Showing empcode, name, deptname, basic, leave type, From date & to date.

Query :

```
CREATE VIEW EMP_LEAVE_VIEW AS
SELECT E.empcode, E.ename, D.dname, E.basic, L.leave_type, L.from_date,
L.to_date
FROM Employee E JOIN Department D ON E.dcode = D.dcode
JOIN LeaveRegister L ON E.empcode = L.empcode;
```

Output :

```
mysql> CREATE VIEW EMP_LEAVE_VIEW AS
-> SELECT E.empcode, E.ename, D.dname, E.basic, L.leave_type, L.from_date, L.to_date
-> FROM Employee E JOIN Department D ON E.dcode = D.dcode
-> JOIN LeaveRegister L ON E.empcode = L.empcode;
Query OK, 0 rows affected (0.03 sec)
```

b) Try to insert a row in the view. Check what happens?

Query :

```
INSERT INTO EMP_LEAVE_VIEW VALUES
(500, 'JOY', 'COMPUTER SCIENCE', 7000, 'CL', '2026-04-01', '2026-04-03');
```

Output :

```
mysql> INSERT INTO EMP_LEAVE_VIEW VALUES
-> (500, 'JOY', 'COMPUTER SCIENCE', 7000, 'CL', '2026-04-01', '2026-04-03');
ERROR 1394 (HY000): Can not insert into join view 'assignments.emp_leave_view' without fields list
mysql>
```

c) Try to increment basic by Rs.100.

Query :

```
UPDATE EMP_LEAVE_VIEW
SET basic = basic + 100;
```

Output :

```
mysql> UPDATE EMP_LEAVE_VIEW
-> SET basic = basic + 100;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

d) Delete the view.

Query :

```
DROP VIEW EMP_LEAVE_VIEW;
```

Output :

```
mysql> DROP VIEW EMP_LEAVE_VIEW;
Query OK, 0 rows affected (0.00 sec)
```


Question 45:

a) Create a table having empcode , Name, deptname, & basic From the existing tables along with the records of the employee who are in a particular department (say, d1) and with a basic Rs. 7000/-

Query :

```
CREATE TABLE D1_EMPLOYEE AS
SELECT E.empcode, E.ename, D.dname, E.basic
FROM Employee E JOIN Department D ON E.dcode = D.dcode
WHERE E.dcode='D1' AND E.basic=7000;
```

Output :

```
mysql> CREATE TABLE D1_EMPLOYEE AS
-> SELECT E.empcode, E.ename, D.dname, E.basic
-> FROM Employee E JOIN Department D ON E.dcode = D.dcode
-> WHERE E.dcode='D1' AND E.basic=7000;
Query OK, 0 rows affected (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

b) From the existing table, add the employees with the basic salary greater than or equal to 7000/-

Query :

```
INSERT INTO D1_EMPLOYEE
SELECT E.empcode, E.ename, D.dname, E.basic
FROM Employee E JOIN Department D ON E.dcode = D.dcode
WHERE E.basic >= 7000;
```

Output :

```
mysql> INSERT INTO D1_EMPLOYEE
-> SELECT E.empcode, E.ename, D.dname, E.basic
-> FROM Employee E JOIN Department D ON E.dcode = D.dcode
-> WHERE E.basic >= 7000;
Query OK, 2 rows affected (0.00 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

c) Alter the table to add a net pay column.

Query :

```
ALTER TABLE D1_EMPLOYEE
ADD netpay DECIMAL(10,2);
```

Output :

```
mysql> ALTER TABLE D1_EMPLOYEE
-> ADD netpay DECIMAL(10,2);
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

d) Replace net pay with 1.5* Basic.

Query :

```
UPDATE D1_EMPLOYEE
SET netpay = basic * 1.5;
```

Output :

```
mysql> UPDATE D1_EMPLOYEE
-> SET netpay = basic * 1.5;
Query OK, 2 rows affected (0.00 sec)
Rows matched: 2 Changed: 2 Warnings: 0
```

```
mysql> SELECT * FROM D1_EMPLOYEE;
+-----+-----+-----+-----+-----+
| empcode | ename | dname          | basic | netpay |
+-----+-----+-----+-----+-----+
| 103     | SUMAN | ELECTRONICS    | 8000.00 | 12000.00 |
| 201     | ANIK  | COMPUTER SCIENCE | 7600.00 | 11400.00 |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

e) Try to remove the net net pay column. [It may require no. of steps]

Query :

```
ALTER TABLE D1_EMPLOYEE
DROP COLUMN netpay;
```

Output :

```
mysql> ALTER TABLE D1_EMPLOYEE
-> DROP COLUMN netpay;
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0
```


Question 46: Drop all the tables that you have created .

Query :

```
DROP TABLE LeaveRegister;
```

```
DROP TABLE Employee;
```

```
DROP TABLE Department;
```

```
DROP TABLE D1_EMPLOYEE;
```

Output :

```
mysql> DROP TABLE LeaveRegister;  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> DROP TABLE Employee;  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> DROP TABLE Department;  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> DROP TABLE D1_EMPLOYEE;  
Query OK, 0 rows affected (0.01 sec)
```

Question 47:

- a) Create EMP table with ECODE(primary key), ENAME, DCODE, GRADE, BASIC & JN-DT as the columns.[except BASIC & JN-DT,all columns are of char type and size of Grade is 1.]
b) Insert number of rows.

Query :

```
CREATE TABLE EMP(
    ECODE CHAR(5) PRIMARY KEY,
    ENAME CHAR(20),
    DCODE CHAR(5),
    GRADE CHAR(1),
    BASIC NUMBER(8,2),
    JN_DT DATE
);

INSERT ALL
    INTO EMP VALUES ('E001','RAHUL','D01','A',25000,TO_DATE('10-JAN-2020','DD-MON-YYYY'))
    INTO EMP VALUES ('E002','AMIT','D01','A',28000,TO_DATE('15-FEB-2021','DD-MON-YYYY'))
    INTO EMP VALUES ('E003','SUMAN','D01','B',22000,TO_DATE('20-MAR-2022','DD-MON-YYYY'))
    INTO EMP VALUES ('E004','RINA','D02','A',30000,TO_DATE('12-APR-2020','DD-MON-YYYY'))
    INTO EMP VALUES ('E005','ANITA','D02','B',24000,TO_DATE('22-MAY-2021','DD-MON-YYYY'))
    INTO EMP VALUES ('E006','ARUN','D02','B',26000,TO_DATE('11-JUN-2022','DD-MON-YYYY'))
    INTO EMP VALUES ('E007','JOY','D03','A',35000,TO_DATE('05-JUL-2019','DD-MON-YYYY'))
    INTO EMP VALUES ('E008','PRIYA','D03','C',20000,TO_DATE('16-AUG-2020','DD-MON-YYYY'))
    INTO EMP VALUES ('E009','SOURAV','D03','C',21000,TO_DATE('25-SEP-2021','DD-MON-YYYY'))
SELECT * FROM DUAL;
```

Output :

```
SQL> CREATE TABLE EMP(
2     ECODE CHAR(5) PRIMARY KEY,
3     ENAME CHAR(20),
4     DCODE CHAR(5),
5     GRADE CHAR(1),
6     BASIC NUMBER(8,2),
7     JN_DT DATE
8 );
```

Table created.

```
SQL> INSERT ALL
2     INTO EMP VALUES ('E001','RAHUL','D01','A',25000,TO_DATE('10-JAN-2020','DD-MON-YYYY'))
3     INTO EMP VALUES ('E002','AMIT','D01','A',28000,TO_DATE('15-FEB-2021','DD-MON-YYYY'))
4     INTO EMP VALUES ('E003','SUMAN','D01','B',22000,TO_DATE('20-MAR-2022','DD-MON-YYYY'))
5     INTO EMP VALUES ('E004','RINA','D02','A',30000,TO_DATE('12-APR-2020','DD-MON-YYYY'))
6     INTO EMP VALUES ('E005','ANITA','D02','B',24000,TO_DATE('22-MAY-2021','DD-MON-YYYY'))
7     INTO EMP VALUES ('E006','ARUN','D02','B',26000,TO_DATE('11-JUN-2022','DD-MON-YYYY'))
8     INTO EMP VALUES ('E007','JOY','D03','A',35000,TO_DATE('05-JUL-2019','DD-MON-YYYY'))
9     INTO EMP VALUES ('E008','PRIYA','D03','C',20000,TO_DATE('16-AUG-2020','DD-MON-YYYY'))
10    INTO EMP VALUES ('E009','SOURAV','D03','C',21000,TO_DATE('25-SEP-2021','DD-MON-YYYY'))
11    SELECT * FROM DUAL;
```

9 rows created.

Question 48 : change the column heading as shown below, So that in subsequent SELECT statement newly set heading will be shown:

ECODE EMPLOYEE CODE

ENAME NAME

DCODE DEPT.CODE

JN-DT JOINING DATE

Query :

```
COLUMN ECODE HEADING 'EMPLOYEE CODE';
```

```
COLUMN ENAME HEADING 'NAME';
```

```
COLUMN DCODE HEADING 'DEPT.CODE';
```

```
COLUMN JN_DT HEADING 'JOINING DATE';
```

```
COLUMN
```

Output :

```
SQL> COLUMN ECODE HEADING 'EMPLOYEE CODE';
```

```
SQL> COLUMN ENAME HEADING 'NAME';
```

```
SQL> COLUMN DCODE HEADING 'DEPT.CODE';
```

```
SQL> COLUMN JN_DT HEADING 'JOINING DATE';
```

```
SQL> COLUMN
```

```
COLUMN    JN_DT ON
```

```
HEADING    'JOINING DATE'
```

```
COLUMN    DCODE ON
```

```
HEADING    'DEPT.CODE'
```

```
COLUMN    ENAME ON
```

```
HEADING    'NAME'
```

```
COLUMN    ECODE ON
```

```
HEADING    'EMPLOYEE CODE'
```

Question 49: Set the format of columns as mentioned below, So that in subsequent SELECT statement ,values appear in the specified format:

***format of BASIC is such that a value of 7000 will be shown as7,000**

***Format of GRADE will be such that full column name appears in the display.**

***For JN-DT format is such that 01-JAN-00 will be shown as JANURY 01,2000**

Query :

```
COLUMN BASIC FORMAT 999,999.99;
```

```
COLUMN GRADE HEADING 'GRADE';
```

```
COLUMN JN_DT FORMAT A20;
```

```
SELECT ECODE,  
       ENAME,  
       DCODE,  
       GRADE,  
       BASIC,  
       TO_CHAR(JN_DT, 'MONTH DD,YYYY') JN_DT  
FROM EMP;
```

Output :

```
SQL> COLUMN BASIC FORMAT 999,999.99;
```

```
SQL> COLUMN GRADE HEADING 'GRADE';
```

```
SQL> COLUMN JN_DT FORMAT A20;
```

```
SQL> SELECT ECODE,  
2          ENAME,  
3          DCODE,  
4          GRADE,  
5          BASIC,  
6          TO_CHAR(JN_DT, 'MONTH DD,YYYY') JN_DT  
7 FROM EMP;
```

EMPLO NAME	DEPT.	G	BASIC	JOINING DATE
E001 RAHUL	D01	A	25,000.00	JANUARY 10,2020
E002 AMIT	D01	A	28,000.00	FEBRUARY 15,2021
E003 SUMAN	D01	B	22,000.00	MARCH 20,2022
E004 RINA	D02	A	30,000.00	APRIL 12,2020
E005 ANITA	D02	B	24,000.00	MAY 22,2021
E006 ARUN	D02	B	26,000.00	JUNE 11,2022
E007 JOY	D03	A	35,000.00	JULY 05,2019
E008 PRIYA	D03	C	20,000.00	AUGUST 16,2020
E009 SOURAV	D03	C	21,000.00	SEPTEMBER 25,2021

9 rows selected.

Question 50:

a) Show the display attributes of all the columns.

Query :

COLUMN;

Output :

```
SQL> COLUMN;
COLUMN      GRADE ON
HEADING      'GRADE'

COLUMN      BASIC ON
FORMAT      999,999.99

COLUMN      JN_DT ON
HEADING      'JOINING DATE'
FORMAT      A20

COLUMN      DCODE ON
HEADING      'DEPT.CODE'

COLUMN      ENAME ON
HEADING      'NAME'

COLUMN      ECODE ON
HEADING      'EMPLOYEE CODE'
```

b) Show the display attributes of particular column.

Query :

COLUMN JN_DT;

Output :

```
SQL> COLUMN JN_DT;
COLUMN      JN_DT ON
HEADING      'JOINING DATE'
FORMAT      A20
```

c) Suppress the newly set attributes of JN-DT .Try a select statement.

Query :

COLUMN JN_DT OFF;

SELECT * FROM EMP;

Output :

```
SQL> COLUMN JN_DT OFF;
SQL> SELECT * FROM EMP;
```

EMPLO	NAME	DEPT.	G	BASIC	JN_DT
E001	RAHUL	D01	A	25,000.00	10-JAN-20
E002	AMIT	D01	A	28,000.00	15-FEB-21
E003	SUMAN	D01	B	22,000.00	20-MAR-22
E004	RINA	D02	A	30,000.00	12-APR-20
E005	ANITA	D02	B	24,000.00	22-MAY-21
E006	ARUN	D02	B	26,000.00	11-JUN-22
E007	JOY	D03	A	35,000.00	05-JUL-19
E008	PRIYA	D03	C	20,000.00	16-AUG-20
E009	SOURAV	D03	C	21,000.00	25-SEP-21

9 rows selected.

d) Reset the newly set attributes of JN-DT

Query :

```
COLUMN JN_DT CLEAR
```

```
COLUMN JN_DT
```

Output :

```
SQL> COLUMN JN_DT CLEAR
```

```
SQL> COLUMN JN_DT
```

```
SP2-0046: COLUMN 'JN_DT' not defined
```

```
SQL> █
```

e) Reset the newly set attributes of all columns.

Query :

```
CLEAR COLUMNS;
```

Output :

```
SQL> CLEAR COLUMNS;
```

```
columns cleared
```

```
SQL> █
```

f) Shown the display attributes of all columns.

Query :

```
COLUMN;
```

Output :

```
SQL> COLUMN;
```

```
SP2-0045: * no COLUMN defined
```

```
SQL> █
```


Question 51:

a) Show the records from EMP table in the ascending order of DCODE . DCODE value will be shown only for the first record of that department (same of DCODE is not repeated)

Query :

```
BREAK ON DCODE;
```

```
SELECT ECODE, ENAME, DCODE, GRADE, BASIC FROM EMP ORDER BY DCODE;
```

Output :

```
SQL> BREAK ON DCODE;
SQL> SELECT ECODE,
  2      ENAME,
  3      DCODE,
  4      GRADE,
  5      BASIC
  6 FROM EMP
  7 ORDER BY DCODE;
```

ECODE	ENAME	DCODE	G	BASIC
E001	RAHUL	D01	A	25000
E002	AMIT		A	28000
E003	SUMAN		B	22000
E004	RINA	D02	A	30000
E005	ANITA		B	24000
E006	ARUN		B	26000
E007	JOY	D03	A	35000
E008	PRIYA		C	20000
E009	SOURAV		C	21000

9 rows selected.

b) Further take measures so that ,after displaying the records of a department it skips one line .

Query :

```
BREAK ON DCODE SKIP 1;
```

```
SELECT ECODE, ENAME, DCODE, GRADE, BASIC FROM EMP ORDER BY DCODE;
```

Output :

```
SQL> BREAK ON DCODE SKIP 1;
SQL> SELECT ECODE,
  2      ENAME,
  3      DCODE,
  4      GRADE,
  5      BASIC
  6 FROM EMP
  7 ORDER BY DCODE;
```

ECODE	ENAME	DCODE	G	BASIC
E001	RAHUL	D01	A	25000
E002	AMIT		A	28000
E003	SUMAN		B	22000
E004	RINA	D02	A	30000
E005	ANITA		B	24000
E006	ARUN		B	26000
E007	JOY	D03	A	35000
E008	PRIYA		C	20000
E009	SOURAV		C	21000

9 rows selected.

c) Further take measures so that records one also ordered on the basis of GRADE within a department & same GRADE value is not repeated.

Query :

```
BREAK ON DCODE SKIP 1 ON GRADE;
SELECT ECODE,
       ENAME,
       DCODE,
       GRADE,
       BASIC
FROM EMP
ORDER BY DCODE, GRADE;
```

Output :

```
SQL> BREAK ON DCODE SKIP 1 ON GRADE;
SQL> SELECT ECODE,
2          ENAME,
3          DCODE,
4          GRADE,
5          BASIC
6 FROM EMP
7 ORDER BY DCODE, GRADE;
```

ECODE	ENAME	DCODE	G	BASIC
E001	RAHUL	D01	A	25000
E002	AMIT			28000
E003	SUMAN		B	22000
E004	RINA	D02	A	30000
E005	ANITA		B	24000
E006	ARUN			26000
E007	JOY	D03	A	35000
E008	PRIYA		C	20000
E009	SOURAV			21000

9 rows selected.

d) Take measure so that at the end of each GRADE in a department it will show average Basic for that grade in that department. At the end of each department, it will show the average & total Basic for the department. At the end of all departments it will show the overall total basic & average basic.

Query :

```
BREAK ON DCODE SKIP 1 ON GRADE SKIP 1;
```

```
COMPUTE AVG OF BASIC ON GRADE;
```

```
COMPUTE AVG OF BASIC ON DCODE;
```

```
COMPUTE SUM OF BASIC ON DCODE;
```

```
COMPUTE AVG OF BASIC ON REPORT;
```

```
COMPUTE SUM OF BASIC ON REPORT;
```

```
SELECT ECODE,
       ENAME,
       DCODE,
```

```

        GRADE,
        BASIC
FROM EMP
ORDER BY DCODE, GRADE;

```

Output :

```

SQL> BREAK ON DCODE SKIP 1 ON GRADE SKIP 1;
SQL> COMPUTE AVG OF BASIC ON GRADE;
SQL> COMPUTE AVG OF BASIC ON DCODE;
SQL> COMPUTE SUM OF BASIC ON DCODE;
SQL> COMPUTE AVG OF BASIC ON REPORT;
SQL> COMPUTE SUM OF BASIC ON REPORT;
SQL> SELECT ECODE,
2      ENAME,
3      DCODE,
4      GRADE,
5      BASIC
6 FROM EMP
7 ORDER BY DCODE, GRADE;

```

ECODE	ENAME	DCODE	G	BASIC

E001	RAHUL	D01	A	25000
E002	AMIT			28000
		*		-----
		a		26500
E003	SUMAN		B	22000
		*		-----
		a		22000
		*****		-----
		sum		75000

ECODE	ENAME	DCODE	G	BASIC

E004	RINA	D02	A	30000
		*		-----
		a		30000
E005	ANITA		B	24000
E006	ARUN			26000
		*		-----
		a		25000

ECODE	ENAME	DCODE	G	BASIC

		sum		80000
E007	JOY	D03	A	35000
		*		-----
		a		35000
E008	PRIYA		C	20000
E009	SOURAV			21000
		*		-----
		a		20500

ECODE	ENAME	DCODE	G	BASIC

		*****		-----
		sum		76000

9 rows selected.

Question 52: Accept an employee code from user . Find the name of the employee from EMP table. If the employee is not existing then indicate it through a suitable message.

Query :

```
SET SERVEROUTPUT ON;
SET VERIFY OFF;
ACCEPT S_ECODE CHAR PROMPT 'Enter Employee Code : '
DECLARE
    V_ENAME EMP.ENAME%TYPE;
BEGIN
    SELECT ENAME INTO V_ENAME FROM EMP WHERE ECODE='&S_ECODE';
    DBMS_OUTPUT.PUT_LINE('Employee Name : '||V_ENAME);
EXCEPTION
    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('NO EMPLOYEE WITH ID : &S_ECODE. ');
    WHEN OTHERS THEN
        DBMS_OUTPUT.PUT_LINE(SQLERRM);
END;
/
```

Code :

```
SQL> @q1.sql
Enter Employee Code : e001
NO EMPLOYEE WITH ID : e001

PL/SQL procedure successfully completed.

SQL> @q1.sql
Enter Employee Code : E001
Employee Name : RAHUL

PL/SQL procedure successfully completed.
```

Question 53: Write a PL / SQL block to add row in EMP table . If employee code is duplicate/dept code is not present in DEPT table then message is to be shown & row can not be added.

Query :

```
SET SERVEROUTPUT ON;
SET VERIFY OFF;
ACCEPT T_ECODE CHAR PROMPT 'Employee Code : '
ACCEPT T_ENAME CHAR PROMPT 'Employee Name : '
ACCEPT T_DCODE CHAR PROMPT 'Department Code : '
ACCEPT T_GRADE CHAR PROMPT 'Grade : '
ACCEPT T_BASIC NUMBER PROMPT 'Basic Salary : '
ACCEPT T_JNDT CHAR PROMPT 'Joining Date (DD-MON-YYYY) : '
DECLARE
    VAR_COUNT NUMBER;
    DUPLICATE_EMP EXCEPTION;
    DEPT_NOT_EXISTS EXCEPTION;
BEGIN
    SELECT COUNT(*) INTO VAR_COUNT FROM EMP WHERE ECODE='&T_ECODE';
    IF VAR_COUNT>0 THEN
        RAISE DUPLICATE_EMP;
    ELSE
        SELECT COUNT(*) INTO VAR_COUNT FROM DEPT WHERE DCODE='&T_DCODE';
        IF VAR_COUNT=0 THEN
            RAISE DEPT_NOT_EXISTS;
        ELSE
            INSERT INTO EMP
VALUES('&T_ECODE', '&T_ENAME', '&T_DCODE', '&T_GRADE', &T_BASIC, TO_DATE('&T_JNDT', '
DD-MON-YYYY'));
            COMMIT;
            DBMS_OUTPUT.PUT_LINE('RECORD INSERTED. ');
        END IF;
    END IF;
EXCEPTION
    WHEN DUPLICATE_EMP THEN
        DBMS_OUTPUT.PUT_LINE('EMPLOYEE EXISTS WITH ECODE : &T_ECODE. ');
    WHEN DEPT_NOT_EXISTS THEN
        DBMS_OUTPUT.PUT_LINE('DEPATMENT DOES NOT EXISTS WITH DCODE :
&T_DCODE. ');
    WHEN OTHERS THEN
        DBMS_OUTPUT.PUT_LINE(SQLERRM);
END;
/
```

Code :

```
SQL> @q2.sql  
Employee Code : E019  
Employee Name : BINOY  
Department Code : D01  
Grade : A  
Basic Salary : 70000  
Joining Date (DD-MON-YYYY) : 12-AUG-2026  
RECORD INSERTED.
```

PL/SQL procedure successfully completed.

```
SQL> @q2.sql  
Employee Code : E019  
Employee Name : BINOY  
Department Code : D01  
Grade : A  
Basic Salary : 70000  
Joining Date (DD-MON-YYYY) : 12-AUG-2026  
EMPLOYEE EXISTS WITH ECODE : E019
```

PL/SQL procedure successfully completed.

Question 54: From the EMP table find the name of top 5 (according to basic salary) employees.

Query :

```
SET SERVEROUTPUT ON;
SET VERIFY OFF;
DECLARE
    VAR_COUNT NUMBER;
BEGIN
    SELECT COUNT(*) INTO VAR_COUNT FROM EMP;
    IF VAR_COUNT<5 THEN
        FOR R IN (SELECT * FROM EMP) LOOP
            DBMS_OUTPUT.PUT_LINE(RPAD(R.ENAME,20)||' '||R.BASIC);
        END LOOP;
        DBMS_OUTPUT.PUT_LINE('TABLE HAVE '||VAR_COUNT||' ROWS.');
```

ELSE

```
        DBMS_OUTPUT.PUT_LINE('TOP 5 EMPLOYEES');
        FOR R IN ( SELECT * FROM ( SELECT ENAME, BASIC FROM EMP ORDER BY BASIC
DESC ) WHERE ROWNUM<=5) LOOP
            DBMS_OUTPUT.PUT_LINE(RPAD(R.ENAME,20)||' '||R.BASIC);
        END LOOP;
    END IF;
EXCEPTION
    WHEN OTHERS THEN
        DBMS_OUTPUT.PUT_LINE(SQLERRM);
END;
/
```

Code :

```
SQL> @q3.sql
TOP 5 EMPLOYEES
BINoy                70000
JOY                   35000
RINA                  30000
AMIT                  28000
ARUN                  26000
```

PL/SQL procedure successfully completed.

Question 55: Accept a department code from the user . Delete all the employee rows with that department code. Show how many rows have been deleted.

Query :

```
SET SERVEROUTPUT ON;
SET VERIFY OFF;
ACCEPT T_DCODE CHAR PROMPT 'Enter Department Code : '
DECLARE
    VAR_COUNT NUMBER;
    NO_EMP_FOUND EXCEPTION;
BEGIN
    DELETE FROM EMP WHERE DCODE='&T_DCODE';
    VAR_COUNT := SQL%ROWCOUNT;
    IF VAR_COUNT = 0 THEN
        RAISE NO_EMP_FOUND;
    END IF;
    COMMIT;
    DBMS_OUTPUT.PUT_LINE(VAR_COUNT || ' row(s) deleted.');
```

EXCEPTION

```
    WHEN NO_EMP_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('NO EMPLOYEE FOUND IN DEPARTMENT &T_DCODE.');
```

WHEN OTHERS THEN

```
        DBMS_OUTPUT.PUT_LINE(SQLERRM);
```

END;

/

Code :

```
SQL> @q4.sql
Enter Department Code : D01
4 row(s) deleted.

PL/SQL procedure successfully completed.
```

```
SQL> @q4.sql
Enter Department Code : D10
NO EMPLOYEE FOUND IN DEPARTMENT D10

PL/SQL procedure successfully completed.
```

Question 56: Consider the following tables:

Table: ORDERMAST

Structure:

ORDER_NO CHAR(5) primary key

ORDER_DT DATE

Table: ORDERDETAILS

Structure:

ORDER_NO CHAR(5)

ITEM_NO CHAR(5)

QTY NUMBER

ORDER_NO + ITEM_NO is primary key

Table: DELIVERYMAST

Structure:

DELV_NO CHAR(5) primary key

ORDER_NO CHAR(5)

DELV_DT DATE

Table: DELIVERY_DETAILS

Structure:

DELV_NO CHAR(5)

ITEM_NO CHAR(5)

QTY NUMBER

DELV_NO+ITEM_NO is primary key

Write a PL/SQL block which accepts two dates & find the information of the pending delivery for the orders placed between the two dates supplied by the user. Note, an order can have request for multiple items. Against an order, multiple delivery with partial shipment is possible.

Query :

SET SERVEROUTPUT ON;

SET VERIFY OFF;

SET LINESIZE 200;

ACCEPT T_FROM CHAR PROMPT 'Enter From Date (DD-MON-YYYY): '

ACCEPT T_TO CHAR PROMPT 'Enter To Date (DD-MON-YYYY): '

DECLARE

BEGIN

DBMS_OUTPUT.PUT_LINE('PENDING DELIVERY REPORT');

DBMS_OUTPUT.PUT_LINE('-----');

FOR R IN

(

SELECT

OD.ORDER_NO,

OD.ITEM_NO,

OD.QTY AS ORDER_QTY,

NVL(SUM(DD.QTY),0) AS DELIVERED_QTY,

OD.QTY - NVL(SUM(DD.QTY),0) AS PENDING_QTY

FROM ORDERMAST OM JOIN ORDERDETAILS OD

ON OM.ORDER_NO = OD.ORDER_NO

LEFT JOIN DELIVERYMAST DM

ON OM.ORDER_NO = DM.ORDER_NO

LEFT JOIN DELIVERY_DETAILS DD

```

        ON DM.DELV_NO = DD.DELV_NO AND OD.ITEM_NO = DD.ITEM_NO
WHERE OM.ORDER_DT BETWEEN
      TO_DATE('&T_FROM','DD-MON-YYYY')
AND TO_DATE('&T_TO','DD-MON-YYYY')
GROUP BY
      OD.ORDER_NO,
      OD.ITEM_NO,
      OD.QTY
ORDER BY
      OD.ORDER_NO,
      OD.ITEM_NO
)
LOOP
      DBMS_OUTPUT.PUT_LINE(
        'ORDER NO : ' || R.ORDER_NO ||
        ' ITEM NO : ' || R.ITEM_NO ||
        ' ORDERED QTY : ' || R.ORDER_QTY ||
        ' DELIVERED QTY : ' || R.DELIVERED_QTY ||
        ' PENDING : ' || R.PENDING_QTY
      );
END LOOP;
END;
/

```

Code :

```

SQL> @assign-5/q5.sql
Enter From Date (DD-MON-YYYY): 01-JAN-2000
Enter To Date (DD-MON-YYYY): 01-JAN-2027
PENDING DELIVERY REPORT

```

```

-----
ORDER NO : 0001  ITEM NAME : KEYBD      ORDERED QTY : 15 DELIVERED QTY : 15 PENDING : 0
ORDER NO : 0001  ITEM NAME : LAPTOP     ORDERED QTY : 10 DELIVERED QTY : 10 PENDING : 0
ORDER NO : 0001  ITEM NAME : MOUSE      ORDERED QTY : 25 DELIVERED QTY : 15 PENDING : 10
ORDER NO : 0002  ITEM NAME : MONTR      ORDERED QTY : 12 DELIVERED QTY : 10 PENDING : 2
ORDER NO : 0002  ITEM NAME : SSD01      ORDERED QTY : 20 DELIVERED QTY : 10 PENDING : 10
ORDER NO : 0003  ITEM NAME : PRNTR      ORDERED QTY : 5 DELIVERED QTY : 5 PENDING : 0
ORDER NO : 0003  ITEM NAME : SCANR      ORDERED QTY : 4 DELIVERED QTY : 2 PENDING : 2
ORDER NO : 0004  ITEM NAME : ROUTR      ORDERED QTY : 8 DELIVERED QTY : 8 PENDING : 0
ORDER NO : 0004  ITEM NAME : WEBBCM     ORDERED QTY : 10 DELIVERED QTY : 5 PENDING : 5
ORDER NO : 0005  ITEM NAME : HEADP      ORDERED QTY : 18 DELIVERED QTY : 10 PENDING : 8
ORDER NO : 0005  ITEM NAME : SPKRS      ORDERED QTY : 6 DELIVERED QTY : 6 PENDING : 0
ORDER NO : 0006  ITEM NAME : PENDR      ORDERED QTY : 30 DELIVERED QTY : 20 PENDING : 10
ORDER NO : 0006  ITEM NAME : PHONE      ORDERED QTY : 14 DELIVERED QTY : 8 PENDING : 6
ORDER NO : 0006  ITEM NAME : TABLE     ORDERED QTY : 7 DELIVERED QTY : 7 PENDING : 0

```

PL/SQL procedure successfully completed.

Question 57: Assume a LEAVE table with emp_no,month,no._of_days.

For each employee find the effective basic for the current month as per the formula given below:

BASIC - (BASIC * no. of leave days in the month)/no. of days in that month.

Query :

```
SET SERVEROUTPUT ON;
```

```
SET VERIFY OFF;
```

```
DECLARE
```

```
VAR_DAYS NUMBER;
```

```
BEGIN
```

```
VAR_DAYS := TO_NUMBER(TO_CHAR(LAST_DAY(SYSDATE), 'DD'));
```

```
DBMS_OUTPUT.PUT_LINE('EFFECTIVE BASIC REPORT');
```

```
DBMS_OUTPUT.PUT_LINE('-----');
```

```
FOR R IN
```

```
(
```

```
SELECT
```

```
E.ECODE,
```

```
E.ENAME,
```

```
E.BASIC,
```

```
NVL(L.NO_OF_DAYS,0) LEAVE_DAYS,
```

```
E.BASIC -(E.BASIC * NVL(L.NO_OF_DAYS,0) / VAR_DAYS) EFFECTIVE_BASIC
```

```
FROM EMP E LEFT JOIN LEAVE_TAB L ON E.ECODE = L.EMP_NO
```

```
AND L.MONTH_NAME = TO_CHAR(SYSDATE, 'MON') ORDER BY E.ECODE
```

```
)
```

```
LOOP
```

```
DBMS_OUTPUT.PUT_LINE(
```

```
R.ECODE||' '||
```

```
RPAD(R.ENAME,20)||' '||
```

```
'EFFECTIVE BASIC = '||
```

```
ROUND(R.EFFECTIVE_BASIC,2)
```

```
);
```

```
END LOOP;
```

```
END;
```

```
/
```

Code :

```
SQL> @q6.sql
```

```
EFFECTIVE BASIC REPORT
```

```
-----
```

E001	RAHUL	BASIC = 25000	EFFECTIVE BASIC = 23333.33
E002	AMIT	BASIC = 28000	EFFECTIVE BASIC = 23333.33
E003	SUMAN	BASIC = 22000	EFFECTIVE BASIC = 19800
E004	RINA	BASIC = 30000	EFFECTIVE BASIC = 29000
E005	ANITA	BASIC = 24000	EFFECTIVE BASIC = 24000
E006	ARUN	BASIC = 26000	EFFECTIVE BASIC = 26000
E007	JOY	BASIC = 35000	EFFECTIVE BASIC = 35000
E008	PRIYA	BASIC = 20000	EFFECTIVE BASIC = 20000
E009	SOURAV	BASIC = 21000	EFFECTIVE BASIC = 21000

PL/SQL procedure successfully completed.